

Impacts of Major Storm Flooding on Harbor Water Quality in New York City using Time Series and Spatial Analysis

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Abstract:

Water quality degradation following major storm events poses significant public health risks in urbanized coastal environments. This study examines the impacts of four storm events on bacterial and chemical contaminant levels in New York City Harbor using water quality data obtained from the NYC Department of Environmental Protection Harbor Water Quality Survey (1909-2024). The four storms analyzed were Hurricane Sandy (2012), Tropical Storm Fay (2020), Hurricane Ida (2021), and Tropical Storm Ophelia (2023), selected to represent a range of storm types and flooding mechanisms. For each event, a ± 14 -day window was applied around the storm landfall date and Wilcoxon rank-sum tests were used to compare pre- and post-storm concentrations of Enterococcus, Fecal Coliform, Turbidity, and Nitrate across 16 statistical comparisons. Station-level spatial maps were produced using Inverse Distance Weighted interpolation in R to visualize the geographic distribution of contaminant change across the harbor. Results indicate that storm flooding mechanism, rather than storm intensity alone, is a primary driver of water quality response. Hurricane Sandy, a surge-dominated event, produced significant post-storm bacterial increases of 814% in Enterococcus and 236% in Fecal Coliform, with Spearman correlation confirming contamination continued rising for days after landfall ($\rho = 0.62$, $p = 0.0003$), suggesting prolonged sewage infrastructure damage. In contrast, rainfall-dominated storms including Hurricane Ida, Tropical Storm Fay, and Tropical Storm Ophelia produced dilution effects or no significant change, despite Ida being a Category 4 hurricane. These findings suggest that flood mechanism is a stronger predictor of harbor water quality response than storm intensity, with direct implications for post-storm public health monitoring and stormwater management in coastal urban systems.

Introduction:

Major storm events in New York City frequently produce intense flooding, especially in low-lying and highly urbanized areas. These floods can overwhelm combined sewer systems and increase pollutants in surrounding waterways, potentially degrading urban water quality. New York City's aging sewer infrastructure is particularly vulnerable to storm surge events, which push ocean water inland and can force raw or partially treated sewage directly into the harbor. This study investigates whether major flood-producing storms lead to measurable changes in water quality contaminants at monitoring sites across NYC Harbor, and whether the type of flooding mechanism, storm surge versus rainfall, determines the direction of that water quality response.

Methodology:

Water quality data were obtained from the NYC Department of Environmental Protection (DEP) Harbor Water Quality Survey, accessed via the NYC Open Data portal (https://data.cityofnewyork.us/Environment/Harbor-Water-Quality/5uug-f49n/about_data). The dataset spans from 1909 to December 2024 and includes measurements collected at multiple monitoring stations across New York City Harbor. Sampling frequency follows a weekly schedule from May through October and monthly from November through April. Four contaminants were selected for analysis: Enterococcus and Fecal Coliform as direct indicators of sewage contamination and public health risk, Turbidity as a measure of physical water disturbance from sediment and debris, and Nitrate as an indicator of fertilizer and nutrient runoff from land.

Four storms that made direct impact on the greater New York City area were selected to represent a range of flooding types: Tropical Storm Fay (July 10, 2020), Tropical Storm Ophelia (September 29, 2023), Hurricane Ida (September 1, 2021), and Hurricane Sandy (October 29, 2012). These events represent three storm categories: tropical storms, hurricane, and catastrophic storm surge, allowing for comparison across different flood-producing mechanisms.

All data processing and statistical analysis were performed in R (R Core Team, 2024) using the tidyverse (Wickham et al., 2019), ggplot2 (Wickham, 2016), and lubridate (Grolemund and Wickham, 2011) packages. Raw contaminant values were converted from character to numeric format prior to analysis. For each storm event, a ± 14 -day window was applied around the storm landfall date to capture water quality conditions both before and after the event, while minimizing the influence of unrelated weather events outside that period. Contaminant observations were aggregated by sampling date, averaging readings across all active monitoring stations, providing a harbor-wide representation of water quality conditions.

Each storm window was divided into two periods: the 14 days before the storm made landfall and the 14 days after. To determine whether water quality changed significantly between these two periods, a Wilcoxon rank-sum test was applied to each storm and contaminant combination, resulting in 16 total statistical comparisons. This test was chosen because water quality measurements are highly variable and do not follow a normal distribution, making it a more reliable option than a standard t-test for this type of data. Results were considered statistically significant at $p < 0.05$.

To investigate whether contaminant levels continued to change in the days following each storm, a Spearman rank correlation was performed between the number of days elapsed since landfall and Enterococcus concentrations for the post-storm period of each event. The Spearman method was used due to the same non-normal distribution considerations as above.

Following initial analysis, an unexpected pattern was observed in which Hurricane Sandy produced bacterial increases after the storm while the three rainfall-dominated storms produced decreases or no significant change. Notably, Hurricane Ida, classified as a Category 4 hurricane, also showed a decrease in bacterial levels following the storm. This suggests that storm intensity alone does not determine the direction of the water quality response. Instead, the primary flooding mechanism appears to play a larger role, as Sandy's impacts on NYC were driven largely by storm surge, whereas Ida, Fay, and Ophelia primarily produced heavy rainfall. This led to an additional comparative analysis examining the average percentage change in bacterial concentrations between pre- and post-storm periods, grouped by storm flooding mechanism: storm surge versus rainfall-dominated events.

Spatial analysis was conducted using R (R Core Team, 2024) with the `sf`, `gstat`, `sp`, `viridis`, `rnatuarearth`, and `ggspatial` packages. Coordinates in the dataset were cleaned and corrected for any swapped latitude and longitude values prior to mapping. Station-level before and after change maps were produced for each storm and contaminant, with Inverse Distance Weighted (IDW) interpolation applied to visualize the spatial distribution of contaminant change across the harbor. Federal Emergency Management Agency (FEMA) National Flood Hazard Layer (NFHL) shapefiles for New York City were obtained from the FEMA Map Service Center (<https://msc.fema.gov>). Monitoring station coordinates were exported from the NYC Department of Environmental Protection (DEP) Harbor Water Quality Survey, available via the NYC Open Data Portal (https://data.cityofnewyork.us/Environment/Harbor-Water-Quality/5uug-f49n/about_data), as point features. A planned spatial join between monitoring stations and FEMA flood zone designations will allow for a future reanalysis of water quality responses grouped by flood zone, providing a more spatially precise assessment of storm impacts across the harbor.

Inverse Distance Weighted (IDW) interpolation was selected due to its simplicity and suitability for datasets with irregularly spaced monitoring stations and moderate spatial density. Unlike geostatistical methods such as kriging, IDW does not require assumptions about spatial autocorrelation structure or variogram modeling, making it appropriate for exploratory spatial analysis where the primary goal is visualization of localized patterns rather than formal spatial prediction. Given the limited number of monitoring stations and uneven spatial distribution across NYC Harbor, IDW provides an interpretable method for representing contaminant gradients.

Results:

Time series plots display contaminant values aggregated by sampling date across all monitoring stations for visualization purposes, while Wilcoxon rank-sum tests were applied to raw individual station observations.

Wilcoxon rank-sum tests comparing pre- and post-storm contaminant concentrations yielded statistically significant results for at least two contaminants in each of the four storm events across 16 total comparisons (Figure 1).

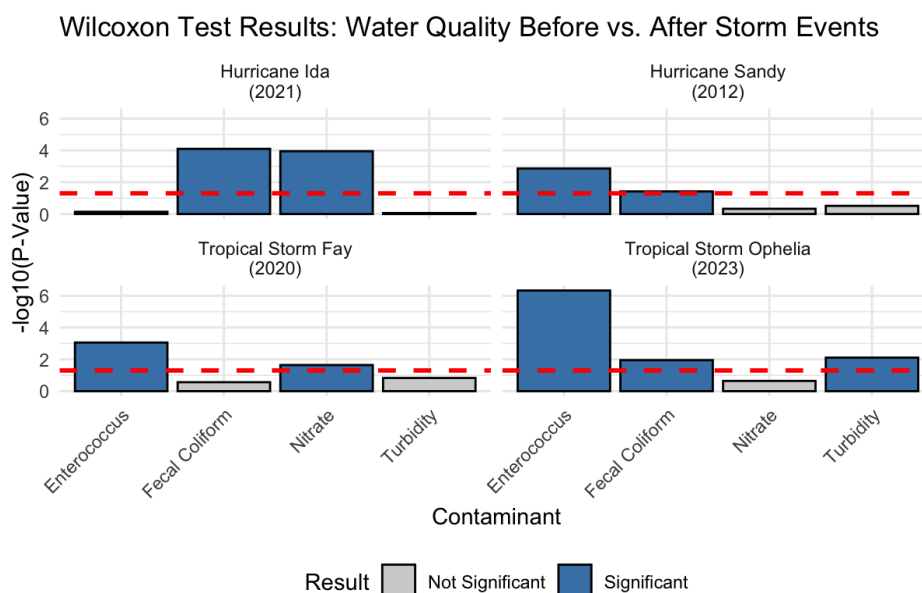


Figure 1. Wilcoxon rank-sum test results for four water quality contaminants across four storm events. Bars represent the negative log10 transformed p-value for each contaminant and storm combination. The red dashed line indicates the significance threshold at $p = 0.05$. Blue bars indicate statistically significant results and gray bars indicate non-significant results. P-values ranged from 4.71×10^{-7} (Ophelia, Enterococcus) to 0.87 (Ida, Turbidity), with 8 of 16 comparisons reaching statistical significance..

For Hurricane Sandy, significant post-storm increases were observed in Enterococcus ($p = 0.0014$) and Fecal Coliform ($p = 0.038$), while Turbidity ($p = 0.31$) and Nitrate ($p = 0.47$) did not reach significance. Average Enterococcus concentrations increased by 814% and Fecal Coliform by 236% in the 14 days following landfall. Spearman correlation confirmed that Enterococcus levels continued to rise with each passing day after Sandy ($\rho = 0.62$, $p = 0.0003$), indicating prolonged rather than immediate contamination.

For Tropical Storm Fay, significant changes were observed in Enterococcus ($p = 0.0009$) and Nitrate ($p = 0.023$), while Fecal Coliform ($p = 0.27$) and Turbidity ($p = 0.15$) did not reach significance. Spearman correlation showed a strong positive trend in Enterococcus in the days following the storm ($\rho = 0.86$, p less than 0.0001), though the overall magnitude of change was substantially smaller than Sandy.

For Hurricane Ida, significant changes were found in Fecal Coliform ($p = 0.00008$) and Nitrate ($p = 0.0001$), while Enterococcus ($p = 0.73$) and Turbidity ($p = 0.87$) showed no significant change. Unlike Sandy, average bacterial concentrations decreased following Ida, with Fecal Coliform declining by 43% and Enterococcus showing no meaningful directional trend (Spearman $\rho = -0.04$, $p = 0.64$).

For Tropical Storm Ophelia, significant changes were observed in Enterococcus ($p = 0.0000005$), Fecal Coliform ($p = 0.011$), and Turbidity ($p = 0.0078$), while Nitrate ($p = 0.23$) did not reach significance. Ophelia produced the strongest dilution response of all four storms, with Enterococcus declining by 87% and Fecal Coliform by 69% following landfall. Spearman correlation confirmed a significant declining trend in Enterococcus in the days after the storm ($\rho = -0.29$, $p = 0.008$).

Comparative analysis of average bacterial percentage change grouped by flooding mechanism revealed a clear contrast between storm types (Figure 2, surge vs. rainfall chart). Hurricane Sandy, the only surge-dominated event, produced large post-storm bacterial increases across both indicators. The three rainfall-dominated storms, Tropical Storm Fay, Hurricane Ida, and Tropical Storm Ophelia, produced either decreases or negligible changes in bacterial concentrations, consistent with a freshwater dilution effect. This pattern held regardless of storm intensity, as Hurricane Ida, a Category 4 hurricane, produced bacterial decreases comparable to the weaker tropical storms.

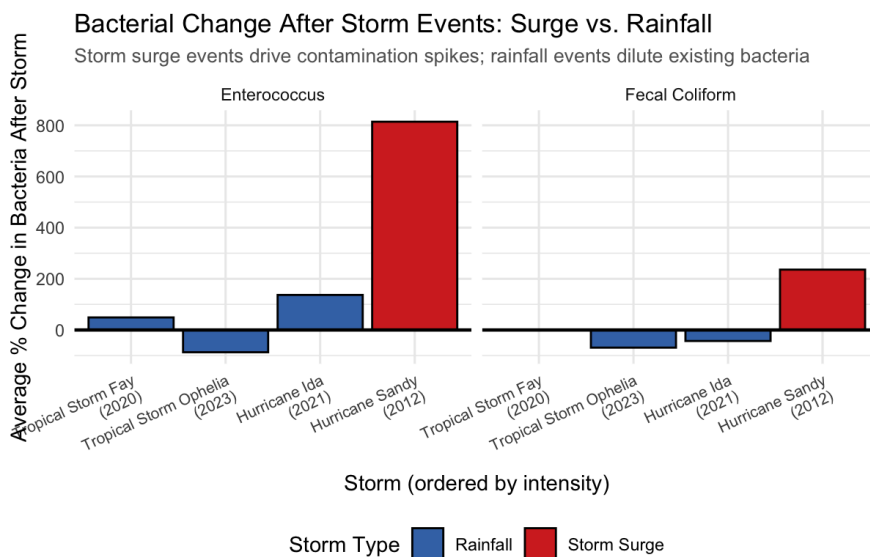


Figure 2. Average percentage change in Enterococcus and Fecal Coliform concentrations in the 14 days following each storm event compared to the 14 days prior, grouped by storm flooding mechanism. Red bars indicate the surge-dominated Hurricane Sandy and blue bars indicate rainfall-dominated storms. Storms are ordered left to right by increasing intensity

Station-level spatial maps produced using IDW interpolation revealed variability in contaminant response across individual monitoring stations within the harbor (Figure 3, Fay Spatial Map). These maps indicate that the harbor-wide averaged results may mask localized contamination hotspots at specific stations, particularly those in closer proximity to flood-prone areas.

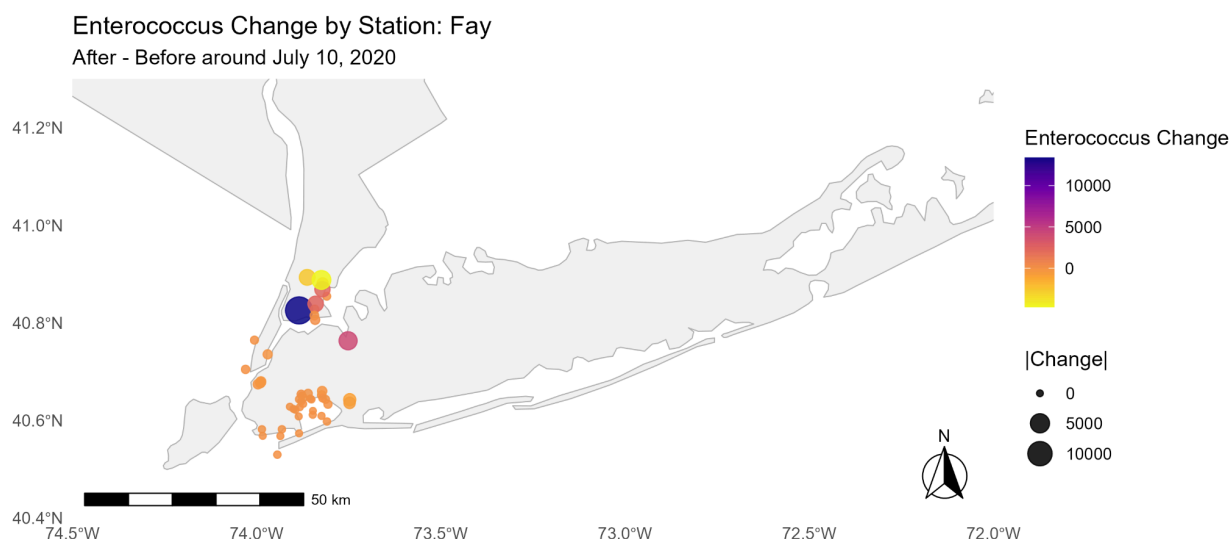


Figure 3. Station-level *Enterococcus* change map for Tropical Storm Fay (July 10, 2020), produced using Inverse Distance Weighted (IDW) interpolation in R. Each point represents an individual NYC DEP monitoring station, with color indicating the direction and magnitude of change in *Enterococcus* concentrations (cells/100mL) between pre- and post-storm periods with warmer colors indicating increases and cooler colors indicating decreases. Dot size reflects the absolute magnitude of change. Spatial maps for remaining storm events are currently being finalized.

Discussion:

This study found that storm flooding mechanism, rather than storm intensity, was the primary driver of water quality change in NYC Harbor. Hurricane Sandy, the only surge-driven storm in this analysis, produced the largest bacterial increases, with *Enterococcus* rising 814% and Fecal Coliform rising 236% in the 14 days following landfall. Spearman correlation confirmed these levels continued rising after the storm ($\rho = 0.62$, $p = 0.0003$), suggesting ongoing sewage system damage rather than a single-day event. In contrast, all three rainfall-dominated storms showed bacterial dilution or no significant change, including Hurricane Ida despite being a Category 4 hurricane, which supports the conclusion that storm category alone is not a reliable predictor of water quality outcomes.

Several limitations should be noted. Daily averages were used for visualization while Wilcoxon tests were applied to raw observations, which creates a mismatch between what is shown and what is tested. Averaging across all stations also removes spatial variability and may mask localized contamination near flood-prone areas. Future analysis incorporating FEMA flood zone designations at the individual station level would strengthen these findings considerably.

Conclusion:

Storm surge events pose a significantly greater risk to harbor water quality than rainfall-dominated storms, regardless of storm intensity. These findings have practical implications for post-storm public health monitoring in coastal cities with aging sewer infrastructure. Future work will integrate FEMA flood zone data with station-level water quality responses to provide a more spatially precise assessment of storm impacts across New York City Harbor.

Credit Authorship Contribution Statement

Carvan, S.: Data collection, statistical analysis, RStudio coding, writing of Abstract, Introduction, Methods, Results, Discussion, Conclusion, and Figures 1, 2, 3; editing.

Carvan, Z.: Data collection, statistical analysis, writing of Methods, Results, and Discussion; data analysis, editing, and contribution to Figure 2.

Marsellos, A.E.: Project supervision, RStudio coding, spatial analysis, review and revision of manuscript, and contribution to Figure 3.

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